Practicum No.9 **Date:24/07/2026**

Reasoning with Forward and Backward Chaining

I. Practical Significance

Forward and backward chaining are inference mechanisms used in rule-based AI systems to derive new knowledge from existing facts. In this practicum, students will implement these reasoning techniques in Prolog to understand data-driven and goal-driven inference, providing a foundation for expert systems, decision support systems, diagnostics, and automation. Industry-relevant job roles include Expert System Developer, AI Reasoning Engineer, Decision Support System Designer, and Knowledge-Based System Analyst.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency.

- ***Use*** the principles of artificial intelligence to solve complex engineering problems.
  (As stated in the Course Structure)

a. ***Apply*** logical reasoning and knowledge representation methods for building rule-based systems and intelligent agents. *(Cognitive Domain)*
b. ***Formulate*** AI-based solutions to real-world problems using appropriate AI techniques. *(Cognitive Domain)*
c. ***Interpret*** AI model outcomes and ensure explainability in decision-making systems. *(Cognitive Domain)*
d. ***Adopt*** ethical principles and responsible practices in the development and deployment of AI systems. *(Affective Domain)*

III. Relevant CO(s) *(As stated in the Course Structure)*

- **CO3. *Develop*** probabilistic and rule-based systems reasoning models for uncertain AI scenarios.

IV. Practical Outcome (PrO) *(As stated in the Course Structure)*

- ***Develop*** Forward and Backward Chaining method for ***the given*** rule-based reasoning problem .

V. Related ADO(s) *(As stated in the Course Structure)*

a) ***Follow*** ethical practices in AI model development and deployment.
b) ***Follow*** adaptability to emerging AI techniques and technologies.
c) ***Follow*** a structured approach to AI driven decision making.
d) ***Follow*** a commitment to continuous learning.

VI. Minimum Underpinning Theory *(Include relevant content & images for this practicum.)*

Forward and Backward Chaining are two core reasoning methods used in rule-based expert systems to draw conclusions from known facts and inference rules. Forward Chaining is data-driven reasoning, where known facts are applied to rules to infer new facts. It proceeds from known facts → conclusions and is ideal for diagnostics and situation assessment. Backward Chaining is goal-driven reasoning, which starts with a goal and works backwards to determine if it can be derived from the known facts using available rules. It is commonly used in systems like Prolog and decision support applications.
In propositional logic terms:
Facts are atomic assertions (e.g., human(Socrates))
Rules are implications (e.g., IF human(x) THEN mortal(x))
Inference uses substitution and unification techniques to match facts with rule premises

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart *(Include probable photos of experimental set-up.)*

Algorithm:

1. Initialize the set of known facts.
2. Select each rule from the knowledge base.

3. Match the rule premises with the known facts.
4. Apply suitable variable substitutions to the matching premises.
5. Infer a new fact using the rule's conclusion.
6. Add the inferred fact to the set of known facts.
7. Repeat Steps 2–6 until no new facts can be inferred.
8. Return the complete set of inferred facts.

VIII. Resources Required

S. No.	Resource	Suggested Broad Specification	Quantity
1	Computer System	Intel i5 or higher, 8 GB RAM, 256 GB SSD, Python 3.x installed	1
2	Prolog Programming Environment	SWI-Prolog, latest stable version	Installed
3	Prolog IDE / Editor	SWI-Prolog IDE or Visual Studio Code with Prolog extension	1

IX. Safety Precautions

a. ***Avoid*** sharing login credentials.
b. ***Ensure*** antivirus software and firewalls are updated.
c. ***Keep*** eatables away from computers.
d. ***Follow*** Institutional laboratory policies.
e. ***Avoid*** storing confidential data in lab computers.

X. Procedure *(Self-Instructional)*

1. ***Interpret*** the flowchart in Section VII to understand forward and backward chaining as reasoning strategies in rule-based systems.
2. ***Define*** a knowledge base in Prolog using facts and rules.
3. ***Represent*** the knowledge using Prolog facts and rules in the form of predicates and clauses.
4. ***Implement*** forward chaining by applying rules to known facts and deriving new facts until no further facts can be inferred.
5. ***Implement*** backward chaining using Prolog queries to recursively verify whether a given goal can be derived from the available facts and rules.
6. ***Execute*** both reasoning approaches using a sample knowledge base such as medical diagnosis, academic eligibility, or traffic management.
7. ***Display*** the inference results and the rules or facts used to derive the conclusions.
8. ***Validate*** the results by checking whether the required goal is successfully proved.
9. Analyze the differences between forward and backward chaining based on the inference process and application.
10. ***Document*** the knowledge base, Prolog queries, inference steps, outputs, and observations in a structured report.
11. ***Submit*** the working Prolog program, queries, outputs, and report for evaluation.

XI. Actual Resources Used (*To be filled by the students*)

S. No	Name of Resource	Version/Configuration	Remarks
1.	Operating System	macOS Tahoe 26.5.2	
2.	Programming Language	Prolog	
3.	Software Development Kit	Prolog	
4.	Libraries	NULL	

XII. Actual Procedure Followed *(To be filled by the students)*

1. The given knowledge base was defined in Prolog using appropriate facts, rules, predicates, and clauses.
2. Forward chaining was performed by applying the rules to known facts and deriving new facts until no further facts could be inferred, while backward chaining was performed by starting with a goal and verifying it using the available facts and rules.
3. The inference results and reasoning steps were executed and verified using different queries to compare the forward and backward chaining processes.

XIII. Observations *(To be filled by the students)*

S. No.	Line Number	Syntax Error	Semantic Error	Exception
1		None	None	None

XIV. Results/Observations *(To be filled by the students)*

Q1.

```
fever(ravi).
cough(ravi).
body_pain(ravi).

respiratory_symptoms(X) :- fever(X), cough(X).
medical_attention(X) :- respiratory_symptoms(X), body_pain(X).
consult_doctor(X) :- medical_attention(X).

diagnose(X) :- fever(X),
        write('Step 1: fever('), write(X), writeln(')'),

        cough(X),
        write('Step 2: cough('), write(X), writeln(')'),

        respiratory_symptoms(X),
        write('Step 3: respiratory_symptoms('), write(X), writeln(')'),

        body_pain(X),
        write('Step 3: body_pain('), write(X), writeln(')'),

        medical_attention(X),
        write('Step 3: medical_attention('), write(X), writeln(')'),

        consult_doctor(X),
        write('Step 3: consult_doctor('), write(X), writeln(')').
```

```
102 ?- consult_doctor(ravi).
true.

103 ?- diagnose(ravi).
Step 1: fever(ravi)
Step 2: cough(ravi)
Step 3: respiratory_symptoms(ravi)
Step 3: body_pain(ravi)
Step 3: medical_attention(ravi)
Step 3: consult_doctor(ravi)
true.
```

Q2.

```
% Facts
ms.
msm.
ssa.
fdl.
olo.

% Rules
msd :- msm.
al :- msd, ssa.
hs :- al, fdl.
sso :- hs, olo.

forward_diagnose :-
    msm,
    write('Step 1: msm'), writeln(''),

    msd,
    write('Step 2: msd'), writeln(''),

    ssa,
    write('Step 3: ssa'), writeln(''),

    al,
    write('Step 4: al'), writeln(''),

    fdl,
    write('Step 5: fdl'), writeln(''),

    hs,
    write('Step 6: hs'), writeln(''),

    olo,
    write('Step 7: olo'), writeln(''),

    sso,
    write('Step 8: sso'), writeln('').
```

```
backward_diagnose :-
    sso,
    write('Step 1: sso'), writeln(''),

    hs,
    write('Step 2: hs'), writeln(''),

    al,
    write('Step 3: al'), writeln(''),

    msd,
    write('Step 4: msd'), writeln(''),

    msm,
    write('Step 5: msm'), writeln(''),

    ssa,
    write('Step 6: ssa'), writeln(''),

    fdl,
    write('Step 7: fdl'), writeln(''),

    olo,
    write('Step 8: olo'), writeln('').
```

```
102 ?- sso.
true.

103 ?- forward_diagnose.
Step 1: msm
Step 2: msd
Step 3: ssa
Step 4: al
Step 5: fdl
Step 6: hs
Step 7: olo
Step 8: sso
true.

104 ?- backward_diagnose.
Step 1: sso
Step 2: hs
Step 3: al
Step 4: msd
Step 5: msm
Step 6: ssa
Step 7: fdl
Step 8: olo
true.
```

XV. Interpretation of Results *(Students to state the meaning of the above-obtained results/observations)*

The results demonstrate that both forward and backward chaining successfully derived conclusions from the given knowledge base. Forward chaining followed a data-driven approach by deriving new facts from existing facts, whereas backward chaining followed a goal-driven approach by working backwards from the required conclusion.

XVI. Conclusions *(Students to draw conclusions (or take decisions) based on the interpretation of results)*

Forward and backward chaining were successfully implemented and tested using Prolog. The experiment demonstrated how facts and rules can be used to perform rule-based inference using both data-driven and goal-driven approaches. The practicum provided an understanding of how chaining methods support reasoning in expert systems and other AI applications.

XVII. Suggested Practicum Related Questions:

Note: *Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes.*

1. What is the difference between forward chaining and backward chaining?
2. In which scenarios would you prefer forward chaining over backward chaining, and vice versa?
3. How are facts and rules represented in your inference engine?
4. What is unification, and why is it important in rule-based reasoning?
5. How does backward chaining implement goal-driven inference?
6. How did your program substitute variables in rules with actual values from facts?
7. What logic did you use to infer new facts in forward chaining?
8. Can your implementation handle multiple rules firing in sequence?
9. How did you manage duplicate or redundant inferences in your engine?
10. How would you extend this system to support predicate logic or multi-step reasoning?
11. What types of domains can benefit from chaining-based reasoning systems?
12. What were the final inferred facts in your forward chaining execution?
13. How would you verify if a goal is provable using backward chaining?
14. What are the limitations of chaining methods in large or uncertain knowledge bases?
15. How would you integrate your inference engine into an expert system?

XVIII References / Suggestions for further reading

Online Tutorials & Articles

a. GeeksforGeeks – Inference Engine in AI (Forward and Backward Chaining- https://www.geeksforgeeks.org/inference-engine-in-artificial-intelligence/
b. TutorialsPoint – Rule-Based Systems https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_rule_based_systems.htm

Interactive Platforms for Practicing

a. Try Python – Online Python Playground - https://trypython.org/
b. Replit – Inference Engine Practice in Python - https://replit.com/

Video Lectures & Courses

a. NPTEL – Artificial Intelligence: Knowledge Representation and Reasoning - https://onlinecourses.nptel.ac.in/noc23-cs88
b. Forward and Backward Chaining - https://www.youtube.com/watch?v=9d0z-ZzFwog
c. (HarvardX) – Knowledge-Based Agents- https://cs50.harvard.edu/ai/2020/

XIX Suggested Assessment Scheme

The performance indicators given serve as a guideline for assessing the *'process-related skills/LOs'* (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and *'product-related skills/LOs'*

Note: The weightage for the ***process-related skills*** and ***product-related skills*** will change depending on the PrO, COs, and the competency related to the concerned course.

Performance Indicators	Max. Marks	Marks Obtained
Process related skills: 18 Marks - 60%		
Understanding of Intelligent agents	3	
Application of suitable AI technique/ algorithm	6	
Followed Coding Standards and Best Practices	6	
Implementation of AI agent correctly	3	
Product related skills: 12 Marks - 40%		
Accuracy of Output/Agent behaviour/ model predictions	3	
Evaluation of performance	3	
Error Handling & Debugging	1.5	
Interpretation of Results and Conclusions	1.5	
Practical related questions	1.5	
Submitting the Report on time	1.5	
Total	**30**	

To be filled by Student:

S.No.	Name of the Student	Register No.
1	**Mithun Chakravarthy C**	**URK25CS1231**

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills (18)	**Product-Related Skills (12)**	**Total (30)**		

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